

Exercise 10

1. Describe the purpose of the feature selection process:
 - (a) Why is feature selection done and what is its goal?
 - (b) Which selection methods were presented in the lecture?
 - (c) When did the presented methods select the final set of features and to which of the three categories of search methods presented in the lecture do they belong?
2. The quality of selected feature subsets is analyzed in the lecture, among other things, regarding inter-feature redundancies. Describe the measure, and explain:
 - (a) To which data types can the measure be applied? How might it need to be modified in order to be applicable?
 - (b) In which value range do possible quality values fall? Which value is ideal, which is catastrophic?
 - (c) Describe how you would use the measure and a set of features to assess whether more features should be removed.
3. How can the quality of selected feature subsets be assessed with regard to a performance criterion? Describe and explain the measure that was presented in the lecture:
 - (a) To which data types can the measure be applied? How might it need to be modified to be applicable?
 - (b) In which value range do possible quality values fall? Which value is ideal, which is catastrophic?
 - (c) Describe how you would use the measure and a set of features to assess whether more features should be removed.
4. Describe filters and wrappers and explain how they work.
5. Explain the advantages and disadvantages of filters and wrappers.